

vLLM Evolution Brief

Official-release evolution through 2026-07-18 · latest stable: v0.25.1

One-line conclusion

vLLM evolved from a paged-KV inference engine into a V1/MRv2-oriented serving platform with broader heterogeneous execution, distributed scaling, and memory-management paths.

Release evolution timeline

Stage	Representative releases	Architectural transition
Foundation	v0.1.0 – v0.3.3	paged KV management, continuous batching, early quantization, Multi-LoRA, prefix caching, and FP8 KV cache.
Production engine	v0.4.0 – v0.6.6.post1	automatic prefix caching, chunked prefill, multimodal serving, multi-step scheduling, and asynchronous output.
V1 transition	v0.7.0 – v0.10.2	V1 alpha/default, torch.compile, DeepSeek MLA/EP/DP, Blackwell, and disaggregated prefill.
V1-only scaling	v0.11.0 – v0.13.0	V0 removal, CUDA graphs, CPU KV offload, Model Runner V2 preview, DBO, and disaggregation.
Async and heterogeneous execution	v0.14.0 – v0.20.2	async scheduling, MRv2 maturation, FA4, hybrid cache, realtime/gRPC APIs, elastic EP, and vLLM IR.
MRv2 platform	v0.21.0 – v0.25.1	MRv2 defaults, Rust frontend, multi-tier KV, DeepEP v2, universal speculative decoding, and legacy PagedAttention implementation removal.

Engine path · scale path · model path

Engine: V1 and Model Runner V2 execution paths, compilation, and CUDA graphs. Scale: TP/DP/EP, context/pipeline parallelism, and prefill/decode disaggregation. Models: dense, MoE, multimodal, speech, and hybrid-state-space serving where official sources identify support.

Key version milestones

Date	Release	Official source
2023-06-20	v0.1.0	https://github.com/vllm-project/vllm/releases/tag/v0.1.0
2023-10-16	v0.2.1	https://github.com/vllm-project/vllm/releases/tag/v0.2.1
2024-02-21	v0.3.2	https://github.com/vllm-project/vllm/releases/tag/v0.3.2
2024-07-23	v0.5.3	https://github.com/vllm-project/vllm/releases/tag/v0.5.3
2024-12-27	v0.6.6	https://github.com/vllm-project/vllm/releases/tag/v0.6.6
2025-04-06	v0.8.3	https://github.com/vllm-project/vllm/releases/tag/v0.8.3
2025-08-20	v0.10.1.1	https://github.com/vllm-project/vllm/releases/tag/v0.10.1.1
2026-01-29	v0.15.0	https://github.com/vllm-project/vllm/releases/tag/v0.15.0
2026-04-27	v0.20.0	https://github.com/vllm-project/vllm/releases/tag/v0.20.0
2026-07-14	v0.25.1	https://github.com/vllm-project/vllm/releases/tag/v0.25.1

Deployment takeaways

Why vLLM: broad serving and scaling roadmap with explicit release-level evolution. What to manage: version cadence, model/hardware compatibility, cache and parallel settings, and migration notes for defaults/deprecations.

Sources: official GitHub Releases and official vLLM documentation. Cutoff 2026-07-18; source mode cache.